

NEUROVISION FACE2EEG

Brief algorithm and system explanation

CORE IDEA

The webcam measures facial dynamics. A trained model uses those dynamics to predict EEG-related signals. The system does not measure EEG through the camera.

1. END-TO-END PIPELINE

Camera frame

- > MediaPipe face landmarks and blendshapes
- > geometric, blink, mouth, pose, motion and action-unit features
- > 64-frame temporal window
- > Transformer encoder
- > predicted EEG waveform, band power, spectrum and uncertainty

Feature windows

- > subject normalization -> PCA
- > Gaussian Mixture Model (GMM)
- > descriptive facial-state cluster probabilities

2. FACIAL FEATURE EXTRACTION

Each frame becomes measurable facial signals:

- Eye Aspect Ratio (EAR), blink events and rolling blink rate.
- Mouth Aspect Ratio (MAR) and jaw opening.
- Head pose: pitch, yaw and roll.
- Landmark velocity, acceleration and movement energy.
- MediaPipe blendshapes mapped to approximate action-unit intensities.

These measurements are grouped into 64-frame temporal windows so the model sees short-term behavior, not only one image.

Implementation: neurovision/preprocessing/facial.py and neurovision/app/live_tab.py.

3. WHY USE A GAUSSIAN MIXTURE MODEL?

The project does not begin with labeled facial-state categories. It discovers recurring patterns in facial behavior. Facial behavior is continuous, and one window can partially resemble more than one state.

GMM is useful because:

- It models each state as a Gaussian distribution in feature space.
- It produces soft probabilities, not only a hard label.
- It can represent clusters with different sizes, orientations and variances.
- It gives useful ambiguity information when a window lies between states.

Before GMM, features are normalized per subject and reduced with PCA. PCA keeps 95 percent of variance. The system tests 2 through 8 components and diagonal or full covariance models, then selects the best combination using BIC.

HOW GMM WORKS: EXPECTATION-MAXIMIZATION

1. Initialize several Gaussian components.
2. Expectation step: calculate each window's probability for every component.
3. Maximization step: update each component mean, covariance and mixture weight.
4. Repeat until the likelihood converges.
5. Select the best model using BIC on training data.

The labels are descriptive suggestions such as frequent blinking, high head motion, variable eye openness, increased mouth opening, or low-motion facial state. They are not emotions, diagnoses or brain states.

4. CLASSIFICATION VERSUS CLUSTERING

There is no conventional supervised emotion or clinical classifier in the live pipeline. No labeled softmax model is trained for those classes.

Clustering: GMM discovers facial behavior groups without class labels.

Cluster assignment: the live window is assigned to the highest-probability component.

Displayed state: a descriptive name is attached after inspecting component feature averages.

Classification: this word may describe the assignment informally, but the method is unsupervised clustering, not supervised classification.

K-Means is also evaluated for comparison, but GMM is primary because it provides probabilities.

5. EEG REGRESSION MODEL

The EEG prediction task is supervised regression. Training data contains synchronized facial windows and EEG targets. The default deep architecture is a Transformer encoder:

- Layer normalization and linear feature projection.
- Positional encoding for temporal order.
- Three Transformer encoder layers with four attention heads.
- Mean pooling across the 64-frame sequence.
- Multiple regression heads.

Outputs:

- Waveform: 128 predicted EEG samples.
- Band power: delta, theta, alpha, beta and gamma values.
- Spectrum: 64 spectral-bin values.
- Uncertainty: variance-related values used to estimate confidence.

6. TRAINING OBJECTIVE

The multitask loss combines:

- Smooth L1 loss for waveform accuracy.
- Fourier-domain loss for spectral similarity.
- Mean squared error for band power.
- Pearson correlation loss for waveform shape.
- Heteroscedastic uncertainty loss for confidence estimation.

Training and validation are subject-independent. Windows from the same subject are kept

out of both splits to reduce subject leakage.

7. ADDITIONAL CLASSICAL REGRESSION

The clustering experiment compares:

- Ridge regression.
- RBF-kernel Support Vector Regression.
- Gradient Boosting regression.

Each model is tested with compact facial features alone and with facial features plus GMM cluster probabilities. This checks whether discovered facial-state information improves EEG prediction.

8. LIMITATIONS

- EEG values in the live dashboard are model predictions, not camera measurements.
- GMM states describe facial behavior and are not emotions or neurological diagnoses.
- Predicted EEG bands must be validated against real EEG recordings.
- Subject-independent evaluation is important because facial and EEG patterns vary across people.

MAIN IMPLEMENTATION AREAS

neurovision/models

neurovision/clustering

neurovision/preprocessing

neurovision/realtime